

# JAVA INTERFACE PROGRAM

Application: Electricity Bill | Program 3

## 1. Aim

To write a Java program using an interface where the House class implements the Electricity interface and calculates the electricity bill based on units consumed and rate per unit.

## 2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named **Program3.java**.
3. Enter the Java source code and save the file.
4. Compile and run the program.
5. Create the Electricity interface with calculateBill().
6. Create the House class implementing Electricity.
7. Assign units consumed and rate per unit.
8. Create a House object and call calculateBill() to display the bill.

## 3. Algorithm

1. Start the program.
2. Create the Electricity interface with calculateBill().
3. Create the House class implementing Electricity.
4. Declare units as 100 and rate as 6.
5. Calculate bill as units  $\times$  rate.
6. Display units consumed, rate per unit, and electricity bill.
7. Create a House object in main().
8. Call calculateBill().
9. Stop the program.

## 4. Explanation of the Program

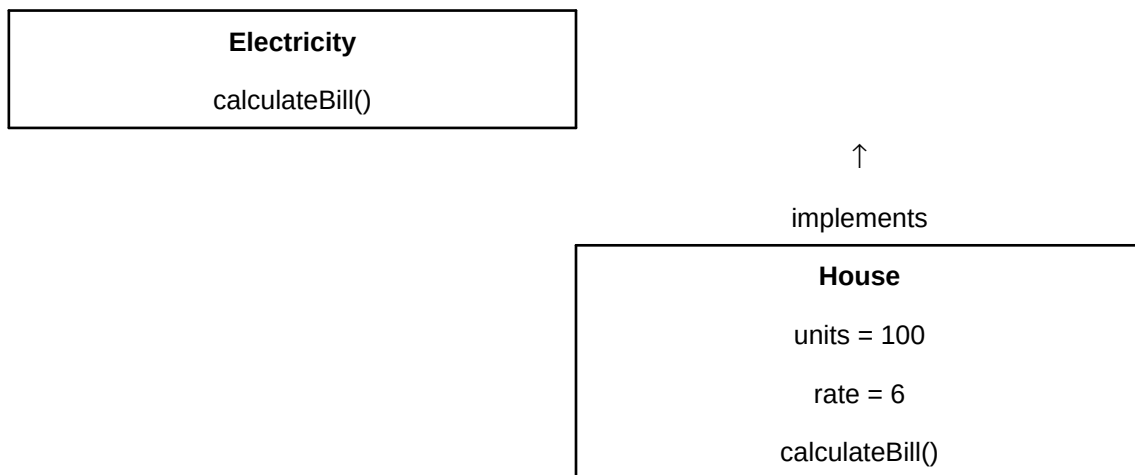
**Electricity interface:** It declares the calculateBill() method that must be implemented by the class using the interface.

**House class:** It implements Electricity and provides the implementation of calculateBill().

**Calculation:** The bill is calculated by multiplying units consumed by the rate per unit.

**House object:** The statement House h = new House(); creates an object and h.calculateBill() executes the bill calculation.

## 5. Interface Implementation Diagram



## 6. Java Source Code

```
// Program 3: C1 implements I1
// Application: Electricity Bill

interface Electricity {
    void calculateBill();
}

class House implements Electricity {

    int units = 100;
    int rate = 6;

    public void calculateBill() {
        int bill = units * rate;
        System.out.println("Units Consumed = " + units);
        System.out.println("Rate per Unit = " + rate);
        System.out.println("Electricity Bill = " + bill);
    }
}

public class Program3 {
    public static void main(String[] args) {

        House h = new House();
        h.calculateBill();
    }
}
```

## 7. Output of the Program

```
Units Consumed = 100  
Rate per Unit = 6  
Electricity Bill = 600
```

## 8. Result

Thus, the Java program successfully demonstrates interface implementation by implementing the Electricity interface in the House class and calculating the electricity bill as 600.